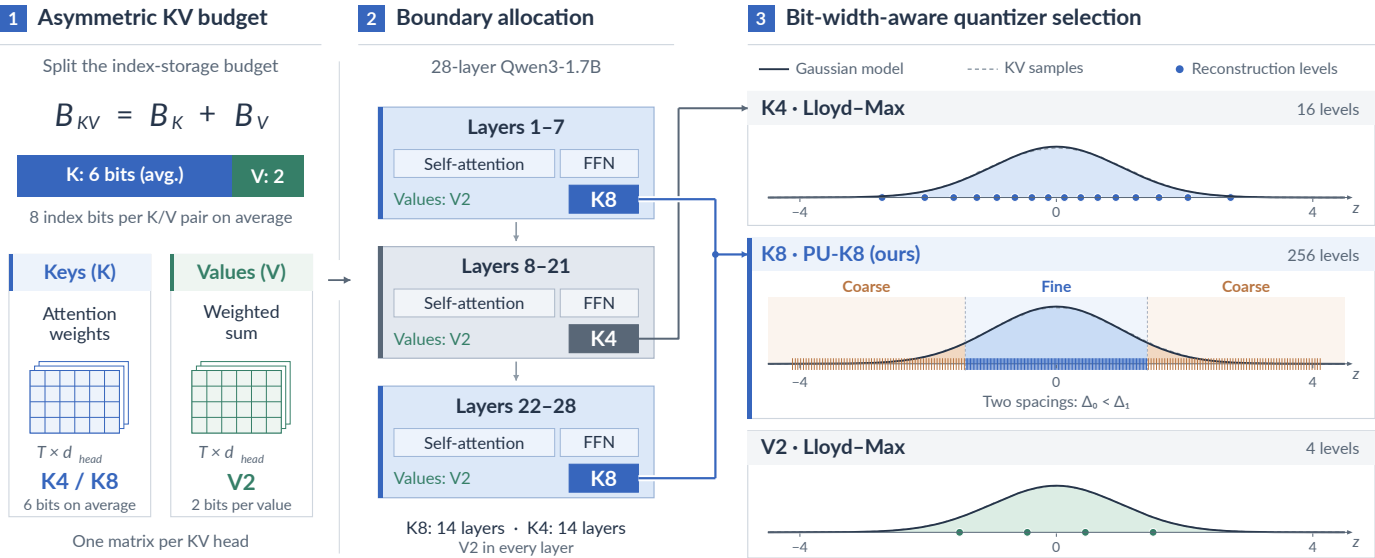

1 Asymmetric KV budget
Split the index-storage budget
$B_{KV} = B_K + B_V$
K: 6 bits (avg.)
V: 2
8 index bits per K/V pair on average
Keys (K)
Attention weights
$T \times d_{head}$
K4 / K8
6 bits on average
Values (V)
Weighted sum
$T \times d_{head}$
V2
2 bits per value
One matrix per KV head
2 Boundary allocation
28-layer Qwen3-1.7B
Layers 1–7
Self-attention
FFN
Values: V2
K8
Layers 8–21
Self-attention
FFN
Values: V2
K4
Layers 22–28
Self-attention
FFN
Values: V2
K8
K8: 14 layers · K4: 14 layers
V2 in every layer
3 Bit-width-aware quantizer selection
Gaussian model
KV samples
Reconstruction levels
K4 · Lloyd–Max
16 levels
−4
0
4
z
K8 · PU-K8 (ours)
256 levels
Coarse
Fine
Coarse
−4
0
4
z
Two spacings: $\Delta_0 < \Delta_1$
V2 · Lloyd–Max
4 levels
−4
0
4
z